

Weihaio Jin

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Education

University of Michigan, Ann Arbor

May 2027

Bachelor of Science in Engineering in Computer Science

Cumulative GPA: 3.82/4.00

Coursework: Machine Learning, NLP, Data Mining, GPU Programming, Operating Systems, Computer Architecture

Skills

Programming Languages: Python, C++, Java, TypeScript, SQL

ML & Research: PyTorch, scikit-learn, OpenCV, Hugging Face, FAISS, NumPy, pandas

Systems & Infrastructure: Docker, AWS, GitHub Actions, CUDA, Redis, PostgreSQL

Publications

Weihaio Jin et al. "VeilLens: Explainable and Granular Analysis and Protection Against Visual Privacy Risks Beyond Visible Appearance." *Manuscript under review*, 2026.

Research Experience

Research Lead | *Ambient Intelligence Lab* | Ann Arbor, Michigan

April 2025 – Present

- Built an end-to-end multimodal privacy-defense pipeline integrating VLM reasoning, open-vocabulary grounding, SAM-based segmentation, diffusion inpainting, and iterative residual-risk verification.
- Designed the interface between language-level privacy reasoning and pixel-level image editing, using structured clue representations and token-aligned filtering to achieve 89% grounding success, versus 64–76% for simpler query formats.
- Developed an evaluation pipeline spanning ~1,500 images and 20,000+ VLM API calls across four frontier attackers, four defense baselines, three privacy tasks, and a cross-dataset geolocation holdout.
- Demonstrated a 48.7% reduction in compositional privacy risk and reduced block-level geolocation success from 42.0% to 12.2%, while preserving stronger perceptual quality than the strongest prior defense.

Work Experience

Research Engineering Intern | *Square Kettle, LLC* | Chicago, Illinois

Aug 2025 – Present

- Implemented the core market-structure algorithm in tick-level event-driven simulation (NautilusTrader), with seeded deterministic runs, per-run manifests, and Plotly/Dash error-analysis tooling.
- Built a leakage-resistant evaluation framework using purged walk-forward splits, hidden holdouts, mandatory baselines, bootstrap confidence intervals, multi-seed variance analysis, and documented ablations.
- Developed an agentic experimentation platform in which Claude Code proposed bounded hypotheses, executed changes in isolated Git worktrees, ran deterministic evaluations, and recorded auditable results across 30+ experiments.
- Deployed reproducible validation workflows using Docker, ECR, AWS Batch, and S3, with live market-data ingestion through Databento, Interactive Brokers, and Redis.

Algorithm Development Intern | *Research Institute of Tsinghua, Pearl River Delta* | Guangzhou, China

May 2024 – Aug 2024

- Reproduced and extended a deep-learning architecture for reconstructing subsurface structures from ground-penetrating radar B-scans, incorporating attention-based components for concrete-defect and rebar localization.
- Diagnosed misalignment between MS-SSIM loss and expert-rated reconstruction quality; implemented and evaluated a composite MS-SSIM, focal, and Dice loss that increased internally rated high-quality outputs from 15% to 30% and acceptable outputs from 55% to 76%.
- Built a gprMax synthetic-data pipeline varying electromagnetic properties, geometry, and noise; supported validation on real concrete and tunnel scans, reducing internally measured false detections by 50%.

Selected Projects

Independent Project | *AI News Podcast Generator*

May 2025 – Sep 2025

Built an end-to-end news-to-podcast pipeline using semantic retrieval, LLM-generated dialogue, multi-voice TTS, and automated factchecking; produced 3–5 minutes episodes in 60–90 seconds.

Project Lead | *PubMedQA with Knowledge Distillation*

Jan 2025 – Apr 2025

Led a four-person team building a PyTorch distillation pipeline from an 8B teacher to a 220M student, improving rationale F1 by 7% and achieving 3–4× faster INT8 inference.